

An Introduction to Regression Models in Python

Python is a very versatile piece of software. It offers many choices for Web applications and can be used in thousands of third party modules. Python is also an effective tool for regression analysis.

I think that machine learning, data mining, chemometrics and statistics are not the same thing but there are aspects that are common to all four topics. For example, a chapter about regression analysis (the techniques for estimating the numeric relationship among data elements) can be found in books on

each of them. Some examples include Reference 1 chapter 6, Reference 2 chapter 13, Reference 3 chapter 2, and Reference 4 chapter 7. In the real world, sometimes there are problems with the fundamentals of linear algebra. There is also the idea that a higher explained variance or a higher correlation means a better model. Or that the only way is the use of a commercial spreadsheet. For these reasons, I think that it's necessary to clarify some things with an introductory article.

From a more technical point of view, for example, if I need to measure the concentration of a certain substance in an underground water sample, I must be capable of measuring a concentration less than 10 times the minimum concentration admitted. For example, if the contamination limit is 1.1 micrograms per litre, I must, at least, be capable of measuring a concentration equal to 0.11 micrograms per litre (these values depend on national laws). With a limit settled by the law, the choice of a regression model becomes even more important. That choice is important also in ion chromatography, because it's better to speak of calibration rather than linearity. The calibration can't always be represented by a regression of the first order, but can be represented sometimes, by a second degree curve. So the concept of linearity should be carefully considered, particularly with regard to the measurement for suppressed conductivity (see Reference 5). This problem is not considered in some methods in which the calibration is valid only if it's linear, discarding *a priori* any other type of model.



The toolbox

This article is based on Mint 18 Xfce, Emacs 24.5, Geany and Anaconda. The last software can be freely downloaded from <https://www.continuum.io/downloads>. Then, from the terminal window, type `bash Anaconda3-4.1.1-Linux-x86.sh` and just follow the instructions on the screen. Because the packages PrettyTable and Seaborn are not available

in the Anaconda distribution, I have easily installed them by typing the following code in a terminal window:

```
cd /home/<your-username>/anaconda_4.1.1/bin
easy_install prettytable
easy_install seaborn
```

My Emacs configuration for Python on Linux is almost the same as on Windows (see the September 2016 issue of OSFY). There is only one difference, because I have replaced the line given below:

```
(add-to-list 'exec-path "C:\\WinPython-32bit-3.4.4.2\\python-3.4.4")
```

... with the following ones:

```
(setq python-shell-interpreter "/home/<your-username>/anaconda_4.1.1/bin/python"
      org-babel-python-command "/home/<your-username>/anaconda_4.1.1/bin/python")
```

With respect to Geany, I have settled only the 'Execute' command `/home/<your-username>/anaconda_4.1.1/bin/python "%f"`

Regression models

The first way to build a polynomial model is by using matrix